

Homogeneous-heterogeneous chemical action and non-Fourier flux theory effects in a flow with carbon nanotubes

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Abstract

A theoretical outline is sketched to figure out the stream of polyvinyl alcohol (PVA)-based CNTs along a moving surface with multiple slip effects on homogeneous and heterogeneous chemical reactions. The nonlinear radiant, heat, and nonuniform heat source/sink is incorporated in the energy equation to discuss the heat transport phenomenon. The novel theory of non-Fourier flux model of heat diffusion is also merged in the energy equation. Two kinds of nanoparticles are considered, namely, single wall carbon nanotubes and multiple wall carbon nanotubes, and are suspended in the working PVA to illustrate the flow behavior. Due to a large variety of applications, PVA is used in papermaking, textiles, and a variety of coatings. The principal structure of mathematical model is based upon the system of differential equations, which has been tackled through Runge–Kutta–Fehlberg method in MATLAB software. Graphical images are made vs different physical parameters which emerged in the model. The heat transfer of the fluid enhances with the increase of nonuniform thermal radiation. The homogeneous and heterogeneous chemical reactions slow down the concentration rate.

KEYWORDS

C-CHFM, CNTs, homogeneous/heterogeneous, nonlinear radiation, non-uniform heat source/sink

1 | INTRODUCTION

In physical systems, transportation of thermal energy is accredited by heat transfer, which has abundant utilization in fields like engineering, and in modern processes of industries, contemporary industrial process that absorb heat transmission in tissues, effectiveness of fuel cell, knowledge in food manufacturing, removing heat from electronic devices, targeting of tumors, and several others. Bicentenary before the most excellent model in heat conduction was proposed by Fourier,¹ which became a benchmark to examine the heat transfer facets in different conditions. But it has a pessimistic phase, that is, a parabolic equation in thermal field is created, which was cited as “Paradox of heat conduction” in literature. To overcome this, in 1948, Cattaneo² remodeled this by sticking the term thermal relaxation time with the movement of heat by the proliferation of hot waves with a firm velocity. Soon after, Christov³ further reshaped the Cattaneo model by restoring the ordinary derivative with Oldroyd’s upper-convected derivatives in order to attain the material-invariant formulation. This recovered replica is eminent as Cattaneo–Christov heat flux model (C-CHFM). The outcomes of C-CHFM for flow problems, which are not compressible, are depicted by Tibulle and Zampoli.⁴ Bala Anki Reddy and Suneetha⁵ contemplated the cause of C-CHF over a stretching surface in the Casson fluid flow. In recent times, many researchers^{6–11} reported on various C-CHF models.

Nanotechnology plays startling applications in various domains of engineering. Nanofluid is one of the important sectors of nanotechnology. Choi¹² introduced the idea of nanofluid by dropping solid nano-sized particles in a host fluid: ethylene glycol, oil, water, and so on. Hayat et al¹³ scrutinized the carbon nanotubes with slip. Carbon-based nanoparticles are divided into tubes, horns, and spheres. Nanoparticles of tube shapes are of two kinds: single-walled and multi-walled, depending on the rolled sheets of graphene. They have many uses in medicine, industry, and agriculture due to their vast surface area.¹⁴ Ul Haq et al^{15,16} assessed the impacts of carbon nanotubes (CNTs) over a stretching surface. Hayat et al^{17,18} focused on the CNT effects on a nonlinear stretching sheet.

Multiple wall carbon nanotubes (MWCNTs) are thermally and chemically more stable when compared to single wall carbon nanotubes (SWCNTs) and can produce massively with low cost. MCWNTs have applications in spray painting and plastic items for automobiles.¹⁹ Particularly, in electronic monitors and health care, SWCNTs are used. Prabavathi et al²⁰ focused on the rates of heat transfer with MWCNTs and SWCNTs and noticed that they are inversely proportional.

In nature, magnetic field exists everywhere, magnetohydrodynamics (MHD) occur whenever conducting fluids are available. MHD has vital applications in several engineering fields: nuclear reactor cooling, controlling the boundary layer in aerodynamics, earth magnetic field, MHD power generators, and so forth. Electric and magnetic fields are induced in a fluid, which is electrically conducting by involving a magnetic field. Several authors^{21–24} examined the MHD effects over a stretching sheet in various types of fluids in different situations.

Radiation does not need any medium to propagate. Electromagnetic field wave is the major reason for radiation heat transmission and conditional on shape, temperature, and so forth. While framing a system in industry with a negligible temperature variation inside a fluid, problems are observed. To avoid this complexity, researchers included an extra parameter nonlinear thermal radiation in place of linear thermal radiation. The differences in temperature of the wall and moving fluid can be noticed by the novel parameter. A brief information on this area is quoted in Nadeem,²⁵ Mohammadein,²⁶ Muhammad,²⁷ and Hayat.²⁸

In dynamics of the thermofluids, the heat production or absorption study plays a key position in the analytical thought of boundary layer in regulating heat transfer. Its significance can be witnessed in the fabrication of plastic sheets, heat exclusion as of nuclear fuel shells, chow stuff cargo space, consent of radioactive squander stuff, and several others. By the application of heat source/sink, there is temperature adjustment inside the boundary layer. By generating heat in the fluid mass, the temperature circulation can be changed, which affects the rate of particle deposition. Such studies have many uses in the designing of biomedical appliances, nuclear reactors, semiconductor wafers, and electronic chips. But, the outcome of nonuniform heat source/sink is fitted in heat shifting studies. Some related articles have been described in Reddy,²⁹ Giri,³⁰ Gireesha,³¹ Ramandevi,³² and Hussaina.³³

In many chemical reacting systems like ignition, catalysis, and biochemical structures, homogeneous and heterogeneous actions occur. Some reactions occur slowly or nothing, apart from the existence of catalyst. The catalyst in a reaction quickens the chemical reaction and is said to be autocatalysis. Homogeneous reaction happens among the similar sector (gases, liquids, and solids), whereas heterogeneous is a two sector or different sectors (solid and gas, solid and liquid). For example, oxidation of iron, fog formation, blast of fireworks, and metabolism of food in body, production of ceramics and polymers, hydrometallurgical industry, dispersion of fog, and scattering of temperature and moisture above the agricultural fields are some of the uses of chemical reactions. In these practices, the molecular distribution of species by way of chemical action inside or at the frontier cannot be taken too lightly. Since there exist two classes of reactions namely homogeneous and heterogeneous. The first reaction takes place evenly in the complete given phase, whereas the later reaction comes about in a controlled region or in the boundary of a phase. These two reactions have complex interaction in preparation and using of reactant species. Usually the rate of reaction is based on the intensity of the mass itself. Merkin³⁴ did the earliest study on chemical reactions. This led so many researchers (Reddy and Suneetha,³⁵ Hayat et al,^{36,37} Gholiniaa et al,³⁸ Hashim et al³⁹) to shed light on these types of reactions.

The above literature survey evidently aids that, no one has attempted to reconnoiter the polyvinyl alcohol (PVA)-based CNTs along a moving surface with multiple slip effects on homogeneous and heterogeneous chemical reactions. By encompassing radiant heat, which is not linear and uneven heat source/sink in energy equation, and momentum and thermal slips in boundary conditions. Elucidations are documented by Runge-Kutta fourth order method. Obtained end results are deliberated with portraits and tables. This investigation is useful in nanotube studies and spread out the former explorations.

2 | PROBLEM DESCRIPTION

Considering a 2D steady, viscous, MHD nanofluid in the presence of carbon-based nanotubes over a moving sheet is exposed in Figure 1. Here we consider PVA as a host fluid, which is also resistant to grease, oils, and solvents. It is flexible but strong, and functions as a high oxygen and aroma barrier. The nanofluid medium is filled with single- and multi-walled CNTs as nanoparticles. The flow of the fluid is commenced by the presence of nonlinear thermal radiation with nonuniform heat source/sink. In this study, C-CHF model is adopted. At this juncture, sheet is considered along the x -axis, which is making a corner to y -axis. A field generated by the magnet B_0 is imposed at an angle 90 degrees to the sheet. Under the above assumptions, the velocity of the surface is expressed as $U_w(x) = e x^m$ (where $e > 0$ is a stable

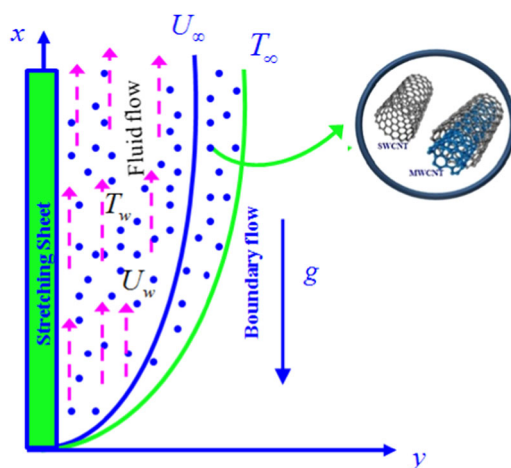
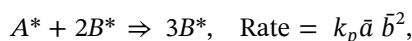


FIGURE 1 Flow framework [Color figure can be viewed at wileyonlinelibrary.com]

one). A^* and B^* are the two chemical varieties taken in the boundary flow for homogeneous and heterogeneous reactions.



where rate constants are précised by k_p and k_q and $\bar{a}$ and $\bar{b}$ are the concentrations of the chemical varieties A^* and B^* . The reactions are assumed isothermal for both processes.

Based on these conventions, the governing flow of the boundary layer equation are stated as:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad (1)$$

$$u \frac{\partial u}{\partial x} + v \frac{\partial v}{\partial y} = \frac{1}{\rho_{nf}} \left(\mu_{nf} \frac{\partial^2 u}{\partial y^2} + (\rho\beta)_f g'(T - T_\infty) - \sigma B_0^2(x)u \right), \quad (2)$$

$$(\rho C_p)_{nf} \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = -\nabla \cdot \bar{q} - \frac{\partial q_r}{\partial y} + \frac{k_f U_w}{xv_f} ((f' A^*(T_w - T_\infty) + B^*(T - T_\infty))), \quad (3)$$

$$u \frac{\partial a_1}{\partial x} + v \frac{\partial a_1}{\partial y} = D_A \frac{\partial^2 a_1}{\partial y^2} - k_p \bar{a} \bar{b}^2, \quad (4)$$

$$u \frac{\partial b_1}{\partial x} + v \frac{\partial b_1}{\partial y} = D_B \frac{\partial^2 b_1}{\partial y^2} + k_p \bar{a} \bar{b}^2, \quad (5)$$

In Equation (3), $A^* > 0$, $B^* > 0$ signifies the heat source and $A^* < 0$, $B^* < 0$, the internal heat sink. In Equations (4) and (5), D_A and D_B are the respective diffusion coefficients of species A^* and B^* .

The C-CHF (q) [3] is expressed as

$$q + \lambda^{**} \left(\frac{\partial q}{\partial t} + \vec{V} \cdot \nabla q - q \cdot \nabla \vec{V} + (\nabla \cdot \vec{V}) q \right) = -k'_f \nabla \cdot T, \quad (6)$$

where λ^{**} is heat flux relaxation time, $\vec{V}$ is velocity vector, and k'_f is the thermal conductivity of the host fluid. At $\lambda^{**} = 0$, Equation (6) is distorted into Fourier's law. The fluid is not compressible so, $\nabla \cdot \vec{V} = 0$ and Equation (6) acquires the following form:

$$q + \lambda^{**} \left(\frac{\partial q}{\partial t} + \vec{V} \cdot \nabla q - q \cdot \nabla \vec{V} \right) = -k'_f \nabla \cdot T. \quad (7)$$

After taking away q from Equations (3) and (7), the equation for energy is

$$\begin{aligned} \frac{\partial T}{\partial x} u + \frac{\partial T}{\partial y} v + \lambda^{**} \left(\frac{\partial T}{\partial x} \left(\frac{\partial u}{\partial x} u + \frac{\partial u}{\partial y} v \right) + \frac{\partial T}{\partial y} \left(\frac{\partial v}{\partial x} u + \frac{\partial v}{\partial y} v \right) + \left(\frac{\partial T}{\partial x} u + \frac{\partial T}{\partial y} v \right)^2 \right) \\ = \alpha_{nf} \frac{\partial^2 T}{\partial y^2} - \frac{1}{(\rho c_p)_{nf}} \frac{\partial q_r}{\partial y} + \frac{1}{(\rho c_p)_{nf}} \{ (T_w - T_\infty) A^* + (T - T_\infty) B^* \}. \end{aligned} \quad (8)$$

Here, α_{nf} is the temperature diffusivity of the nanofluid. By applying Rosseland radiant heat flux,²⁹

$$q_r = -(4/(3K^*)) \text{grad}(e_b). \quad (9)$$

T^4 can be expressed as a linear temperature function and T^4 can be enlarged as Taylor series and accurated after taking out of higher order terms.

$$T^4 = \{1 + (\theta_w - 1)\theta\}^4 T_\infty^4, \quad (10)$$

where, $\theta_w =$ ratio of T_w and T_∞ , and $\theta_w > 1$ is the wall temperature ratio parameter. Thermal conductivities of CNTs can be predicted by many models such as Maxwell, Jeffery, Hamilton, and Crosser. But, the Xue model predicts that these models are exclusively used for particles with minor axial ratio and are of spherical or elliptical in shape, and do not meet the effect of space allocation on CNTs. So, he inaugurated a model pedestal on Maxwell theory on the topic of the nanotubes of rotational elliptical having main axial ratio and compensate the consequence of the space allocation on CNTs. Here, Xue model is operated as follows:

$$\frac{k_{nf}}{k_f} = \frac{(\chi^*) + 2\chi \left(\frac{k_{CNT}}{k_{CNT} - k_f} \right) \log \left(\frac{k_{CNT} + k_f}{2k_f} \right)}{(\chi^*) + 2\chi \left(\frac{k_f}{k_{CNT} - k_f} \right) \log \left(\frac{k_{CNT} + k_f}{2k_f} \right)}, \quad \chi^* = 1 - \chi, \quad (11)$$

where

$$\left. \begin{aligned} \rho_{nf} v_{nf} &= \mu_{nf}, \quad \chi^* = 1 - \chi, \quad \frac{\mu_{nf}}{\mu_f} = (\chi^*)^{-2.5}, \quad \frac{\rho_{nf}}{\rho_f} = (\chi^*) + \chi \frac{\rho_{CNT}}{\rho_f}, \quad \alpha_{nf} = \frac{k_{nf}}{(\rho C_p)_{nf}}, \\ (\rho C_p)_{nf} &= (\chi^*)(\rho C_p)_f + (1 - \chi^*)(\rho C_p)_{CNT}, \quad (\rho \beta)_{nf} = (\chi^*)(\rho \beta)_f + (1 - \chi^*)(\rho \beta)_{CNT}, \end{aligned} \right\} \quad (12)$$

Subject to the boundary conditions:

$$\begin{aligned} \frac{\partial \psi}{\partial y} &= ax^m + l'_1 \frac{\partial^2 \psi}{\partial y^2}, \quad \frac{\partial \psi}{\partial x} = 0, \quad T = T_w + l'_2 \frac{\partial T}{\partial y}, \\ D_A \frac{\partial \bar{a}}{\partial y} &= -D_B \frac{\partial \bar{b}}{\partial y} = k_s \bar{a}, \quad \text{at } y = 0 \\ \frac{\partial \psi}{\partial y} &= 0, \quad T \rightarrow T_\infty, \quad \bar{a} = a_\infty, \quad \bar{b} = 0 \quad \text{as } y \rightarrow \infty, \end{aligned} \quad (13)$$

where $l'_1 = l_1 x^{-\frac{m-1}{2}}$, $l'_2 = l_2 x^{-\frac{m-1}{2}}$ are momentum, thermal slip factors.

Using the similarity transformations,

$$\begin{aligned} \eta &= y \sqrt{\frac{a x^{m-1}}{v_f}}, \quad \psi(x, y) = f(\eta) \sqrt{a x^{m+1} v_f}, \quad \phi(\eta) = \frac{\bar{a}}{a_\infty}, \quad h(\eta) = \frac{\bar{b}}{a_\infty}, \\ u &= a x^m f'(\eta), \quad v = -\frac{1}{2} \{(m+1)f(\eta) + (m-1)\eta f'(\eta)\} \sqrt{a v_f x^{m-1}}, \\ \theta(\eta) &= \frac{T - T_\infty}{T_w - T_\infty}. \end{aligned} \quad (14)$$

Adopting Equations (12) to (14) in Equations (2), (8), (4), and (5), we get

$$\begin{aligned} f''' + \left\{ \left(1 - \chi + \chi \frac{\rho_{CNT}}{\rho_f} \right) \left[\frac{m+1}{2} f f'' - m f'^2 \right] \right\} (1 - \chi)^{2.5} \\ + \left\{ \left(1 - \chi + \chi \frac{(\rho \beta)_{CNT}}{(\rho \beta)_f} \right) \lambda \theta - M^2 f' \right\} (1 - \chi)^{2.5} = 0, \end{aligned} \quad (15)$$

$$\begin{aligned} \frac{1}{pr} \left\{ \theta'' \left\{ \frac{k_{nf}}{k_f} + [1 + (\theta_w - 1)\theta]^3 \frac{4}{3} Rd \right\} + \theta'^2 [1 + (\theta_w - 1)\theta]^2 (\theta_w - 1) 4Rd \right. \\ \left. + (A^* f' + B^* \theta) \right\} \\ + \left(1 - \chi + \chi \frac{(\rho C_p)_{CNT}}{(\rho C_p)_f} \right) \left(\frac{m+1}{2} f \theta' - \gamma \left(\frac{(m+1)(3-m)}{4} f f' \theta' + \frac{(m+1)^2}{4} f^2 \theta'' \right) \right) = 0, \end{aligned} \quad (16)$$

$$\frac{1}{Sc}\phi'' + \frac{m+1}{2}f\phi' - K\phi h^2 = 0, \quad (17)$$

$$\frac{\varepsilon}{Sc}h'' + \frac{m+1}{2}f h' + K\phi h^2 = 0, \quad (18)$$

Together with the conditions at boundary

$$\begin{aligned} f(0) &= 0, \quad f'(0) = 1 + \xi f''(0), \quad \theta(0) = 1 + \zeta \theta'(0), \\ \phi'(0) &= Ks \phi(0), \quad \varepsilon h'(0) = -Ks \phi(0) \quad \text{at } \eta = 0, \\ f'(\infty) &= 0, \quad \theta(\infty) = 0, \quad \phi(\infty) = 1, \quad h(\infty) = 0 \quad \text{as } \eta \rightarrow \infty. \end{aligned} \quad (19)$$

where about

$$\begin{aligned} Sc &= \frac{\nu}{D_A}, \quad \lambda = \frac{Gr_x}{Re_x^2}, \quad Gr_x = \frac{(\rho\beta)_f g (T_w - T_\infty)x^3}{\nu_f^2 \rho_f}, \quad Re_x^2 = \frac{U_w^2 x^2}{\nu_f^2}, \quad \varepsilon = \frac{D_B}{D_A}, \\ M &= B_0 \sqrt{\frac{\sigma}{a\rho_f}} x, \quad \gamma = \lambda^* a x^{m-1}, \quad Rd = \frac{4\sigma^* T_\infty^3}{k k_f}, \quad pr = \frac{\mu_f (\rho C_p)_f}{\rho_f k_f}, \\ Ks &= \frac{k_s}{D_A} \left(\frac{a x^{m-1}}{\nu_f} \right)^{-1/2}, \quad K = \frac{x a_{1\infty}^2 k_c}{U}, \quad \xi = l_1 \sqrt{\frac{a}{\nu_f}}, \quad \zeta = l_2 \sqrt{\frac{a}{\nu_f}}. \end{aligned} \quad (20)$$

Thus $\varepsilon = 1$ for the coefficient of diffusion D_A and D_B is equal.

$$\phi(\eta) + h(\eta) = 1. \quad (21)$$

Equations (17) and (18) under this assumption are reduced to

$$\frac{1}{Sc}\phi'' + \frac{m+1}{2}f\phi' - K\phi(1-\phi)^2 = 0. \quad (22)$$

Related to the boundary conditions

$$\phi'(0) = Ks\phi(0), \quad \text{at } \eta = 0, \quad \phi(\infty) = 1, \quad \text{as } \eta \rightarrow \infty. \quad (23)$$

The shear stress rate and local convectional transmission coefficients are the quantities of interest considered:

$$C_f = \frac{\tau_w}{\rho_f U_w^2}, \quad Nu = \frac{q_w x}{k_f (T_w - T_\infty)}, \quad (24)$$

where τ_w is wall skin friction and q_w is wall heat flux is arranged as at $y = 0$

$$\tau_w = \mu_{nf} \left(\frac{\partial u}{\partial y} \right), \quad q_w = -K_{nf} \left(\frac{\partial T}{\partial y} \right) + (q_r). \quad (25)$$

The reduced shear stress rate and local convectional transmission coefficients can be expressed as

$$C_f = \frac{1}{(1-\chi)^{2.5}} \text{Re}_x^{-\frac{1}{2}} f''(0), \quad Nu = -(\text{Re}_x)^{\frac{1}{2}} \theta'(0) \left\{ \frac{k_{nf}}{k_f} + [1 + (\theta_w - 1)\theta(0)]^3 \frac{4}{3} Rd \right\}. \quad (26)$$

3 | NUMERICAL EXPERIMENTATION

Let,

$$\left. \begin{aligned} A_1 &= [\chi^*]^{2.5} \left((\chi^*) + \chi \frac{\rho_{CNT}}{\rho_f} \right), \quad A_2 = (\chi^*)^{2.5} \left((\chi^*) + \chi \frac{(\rho\beta)_{CNT}}{(\rho\beta)_f} \right), \\ A_3 &= (\chi^*)^{2.5}, \quad A_4 = \left((\chi^*) + \chi \frac{(\rho C_P)_{CNT}}{(\rho C_P)_f} \right), \quad A_5 = \frac{k_{nf}}{k_f}, \quad \chi^* = 1 - \chi \end{aligned} \right\} \quad (27)$$

As the consequential partial differential expressions of this problem are transformed into ordinary differential equations. But these ordinary differential equations (Equations (15), (16), and (22)) are highly nonlinear. We work out this crisis by shooting technique with the Runge–Kutta–Fehlberg (RKF) method. Predominantly, the set of coupled nonlinear ODEs are renewed in to system of first order differential equations as given below:

$$\begin{aligned} f &= p_1, \quad f' = p_2, \quad f'' = p_3, \quad f''' = p'_3, \quad \theta = p_4, \quad \theta' = p_5, \\ \theta'' &= p'_5, \quad \phi = p_6, \quad \phi' = p_7, \quad \phi'' = p'_7, \end{aligned} \quad (28)$$

$$p'_3 = \frac{A_3 M^2 p_2 - A_2 \lambda p_4 - A_1 \left(\frac{m+1}{2} p_1 p_2 - m p_2^2 - \alpha \left(m(m-1) p_2^3 - \frac{m^2-1}{4} \eta p_2^2 p_3 - \frac{(m+1)(3m-1)}{2} p_1 p_2 p_3 \right) \right)}{\left(1 - \alpha A_1 \frac{(m+1)^2}{4} p_1^2 \right)}, \quad (29)$$

$$p'_5 = \frac{4Rd [(\theta_w - 1)p_4 + 1]^2 (\theta_w - 1) p_5^2 + (A^* p_2 + B^* p_4) - Pr A_4 \left(\frac{m+1}{2} p_1 p_5 - \frac{(m+1)(3-m)}{4} p_1 p_2 p_5 \right)}{\left(A_5 + \frac{4}{3} Rd [(\theta_w - 1)p_4 + 1]^3 - \gamma \frac{(m+1)^2}{4} p_1^2 Pr \right)}, \quad (30)$$

$$p'_7 = Sc K p_6 (1 - p_6)^2 - Sc \frac{m+1}{2} p_1 p_7. \quad (31)$$

Associated boundary conditions are

$$\begin{aligned} p_1(0) &= 0, \quad p_2(0) = 1 + \xi p_3(0), \quad p_4(0) = 1 + \zeta p_5(0), \\ p_7(0) &= Ksp_6(0), \quad p_2(\infty) = 0, \quad p_4(\infty) = 0, \quad p_7(\infty) = 1. \end{aligned} \quad (32)$$

To solve the various problems related to boundary layer flows, this technique is successfully used. Insert two initial guesses to $f''(0)$ and $\theta'(0)$ for approximate solution. Here, the step size and convergence criteria are chosen to be 0.001 and 10^{-6} (in all cases).

4 | RESULTS PROCLAMATION

This portion is captivated to figure out the complete understanding of involving parameter on flow, temperature and nanoparticle concentration, the shear stress rate coefficient, and local convectional transmission coefficient files over a stretching sheet. The related parameters are magnetic flux M , thermal buoyancy λ , nonlinear thermal radiation Rd , Prandtl number Pr , Schmidt number Sc , the temperature ratio parameter θ_w , thermal relaxation time γ , volume fraction of the CNTs χ heat source A^* , heat sink B^* , strength coefficient of homogenous reaction K , and Ks is the strength parameter due to heterogeneous reaction. Single- and multi-walled CNTs for PVA (host fluid) are examined here. The numerical standards of CNTs and PVA are shown in Table 1. The numerical calculations have been done for all the physical parameters by assuming the default values $M = 0.2$, $\lambda = 0.1$, $m = 0.5$, $\eta = 0.5$, $Rd = 10.0$, $Pr = 0.72$, $\gamma = 0.2$, $Sc = 0.8$, $\theta_w = 1.1$, $A^* = 0.05$, $B^* = 0.05$, $K = 0.1$, $Ks = 10.0$, $\chi = 0.06$, until otherwise specified. From Table 2, it is clear that the shear stress rate coefficient is an increasing function of Rd and decreasing function of M , Pr , γ , θ_w , A^* , and B^* . All the values of shear stress rate coefficient are negative. From Table 3, it is clear that the local convectional transmission coefficient is an increasing function of A^* , B^* , γ , θ_w , Rd , Pr , and declining function of M . The solid (green colored) line indicates the contour of MWCNTs and dashed (blue coloured) line indicates the contour of SWCNTs. In this segment, the effects of dimensionless parameters are demonstrated in Figures 2 to 17 for the different values of the pertinent parameters.

4.1 | Effect of magnetic parameter (M)

The impact of M on momentum is shown in Figure 2. An increment in magnetic field parameter enhances the resistive type force, branded as Lorentz force is the reason behind it.

TABLE 1 Thermophysical characteristics of base fluids and carbon nanotubes

Physical properties	Primary fluid (polyvinyl alcohol)	Nanoparticles (SWCNT)	Nanoparticles (MWCNT)
C_p (J/kg K)	2000	425	796
ρ (kg/m ³)	1020	2600	1600
κ (W/m K)	0.2	6600	3000
$\beta \times 10^{-5}$ (K ⁻¹)	280.15	2.6	2.8

Abbreviations: MWCNT, multiple wall carbon nanotube; SWCNT, single wall carbon nanotube.

TABLE 2 The values of shear stress rate coefficient and various dimensionless parameters

M	Rd	Pr	γ	θ_w	A^*	B^*	$f''(0)$	
							SWCNT	MWCNT
0.2	10.0	7.2	0.2	1.1	0.05	0.05	-0.5302	-0.5201
0.4	10.0	7.2	0.2	1.1	0.05	0.05	-0.5738	-0.5651
0.6	10.0	7.2	0.2	1.1	0.05	0.05	-0.6372	-0.6302
0.2	11.0	7.2	0.2	1.1	0.05	0.05	-0.5285	-0.5183
0.2	12.0	7.2	0.2	1.1	0.05	0.05	-0.5269	-0.5152
0.2	10.0	8.2	0.2	1.1	0.05	0.05	-0.5330	-0.5230
0.2	10.0	9.2	0.2	1.1	0.05	0.05	-0.5355	-0.5255
0.2	10.0	7.2	0.6	1.1	0.05	0.05	-0.5314	-0.5214
0.2	10.0	7.2	1.0	1.1	0.05	0.05	-0.5328	-0.5229
0.2	10.0	7.2	0.2	1.2	0.05	0.05	-0.5312	-0.5212
0.2	10.0	7.2	0.2	1.3	0.05	0.05	-0.5323	-0.5223
0.2	10.0	7.2	0.2	1.1	0.30	0.05	-0.5315	-0.5214
0.2	10.0	7.2	0.2	1.1	0.55	0.05	-0.5327	-0.5227
0.2	10.0	7.2	0.2	1.1	0.05	0.30	-0.5324	-0.5223
0.2	10.0	7.2	0.2	1.1	0.05	0.55	-0.5344	-0.5243

Abbreviations: MWCNT, multiple wall carbon nanotube; SWCNT, single wall carbon nanotube.

TABLE 3 Deviations in the rate of local convectational transmission coefficient and various dimensionless parameters

M	Rd	Pr	γ	θ_w	A^*	B^*	$-\theta'(0)$	
							SWCNT	MWCNT
0.2	10.0	7.2	0.2	1.1	0.05	0.05	2.0983	2.3421
0.4	10.0	7.2	0.2	1.1	0.05	0.05	1.9321	2.1679
0.6	10.0	7.2	0.2	1.1	0.05	0.05	1.6918	1.9618
0.2	11.0	7.2	0.2	1.1	0.05	0.05	2.3586	2.6045
0.2	12.0	7.2	0.2	1.1	0.05	0.05	2.6085	3.0992
0.2	10.0	8.2	0.2	1.1	0.05	0.05	2.3900	2.6391
0.2	10.0	9.2	0.2	1.1	0.05	0.05	2.6660	2.9201
0.2	10.0	7.2	0.6	1.1	0.05	0.05	2.1154	2.3615
0.2	10.0	7.2	1.0	1.1	0.05	0.05	2.1459	2.3953
0.2	10.0	7.2	0.2	1.2	0.05	0.05	3.5399	3.8076
0.2	10.0	7.2	0.2	1.3	0.05	0.05	5.2930	5.5881
0.2	10.0	7.2	0.2	1.1	0.30	0.05	2.3028	2.5514
0.2	10.0	7.2	0.2	1.1	0.55	0.05	2.5057	2.7593
0.2	10.0	7.2	0.2	1.1	0.05	0.30	2.4148	2.6530
0.2	10.0	7.2	0.2	1.1	0.05	0.55	2.7022	2.9360

Abbreviations: MWCNT, multiple wall carbon nanotube; SWCNT, single wall carbon nanotube.

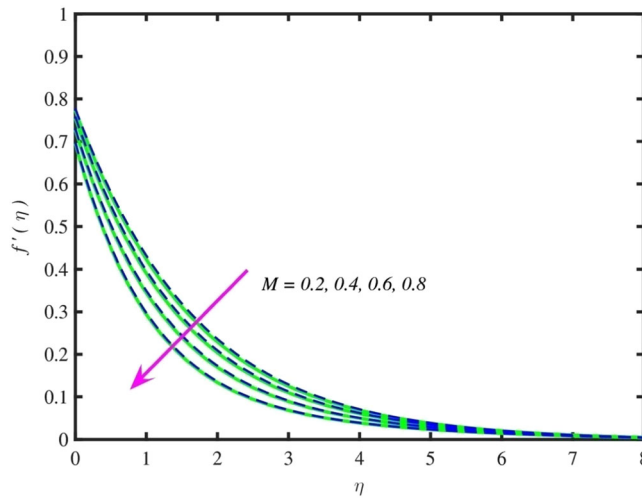


FIGURE 2 Variations of M on velocity [Color figure can be viewed at wileyonlinelibrary.com]

This force boost up the colloid nature of the particles which slow down the liquid velocity. Furthermore, the momentum layer at boundary declines by this force. Velocity profiles overlie for both SWCNT and MWCNT.

4.2 | Effect of momentum (hydrodynamic) slip parameter (ξ)

Figure 3 illustrates the impact of ξ on the velocity distribution. With escalating (ξ) the polymer slips, that is, shears further easily in opposition to the stretching plate surface. This uplift momentum in the boundary of the sheet layer and speed up the flow (Figure 3). The velocity slip effect is strongest at the plate surface ($\eta = 0$). In addition, the velocity profile is elevated for SWCNT.

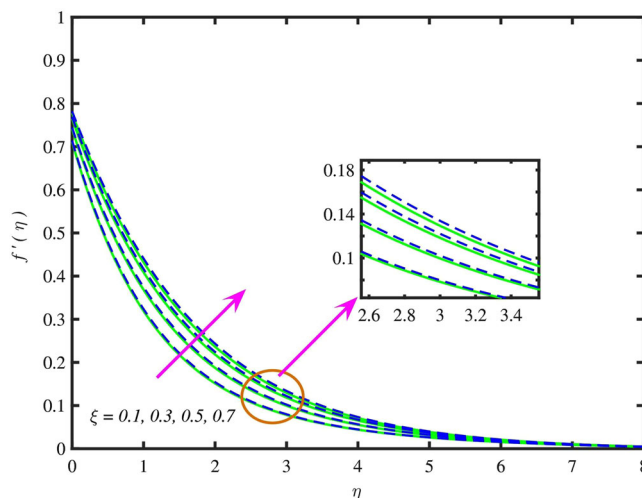


FIGURE 3 Variations of ξ on velocity [Color figure can be viewed at wileyonlinelibrary.com]

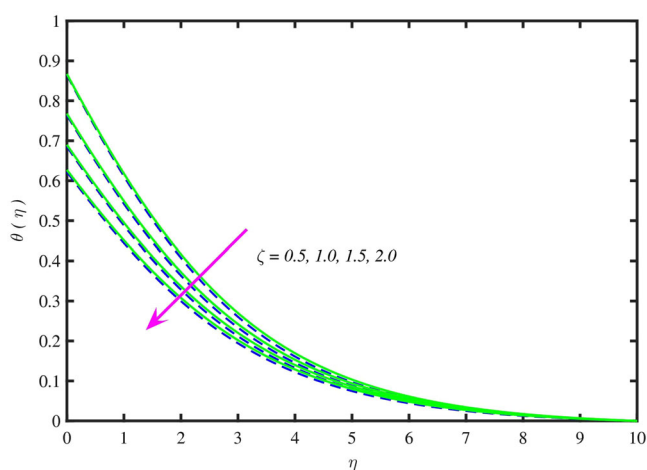


FIGURE 4 Variations of ζ on temperature [Color figure can be viewed at wileyonlinelibrary.com]

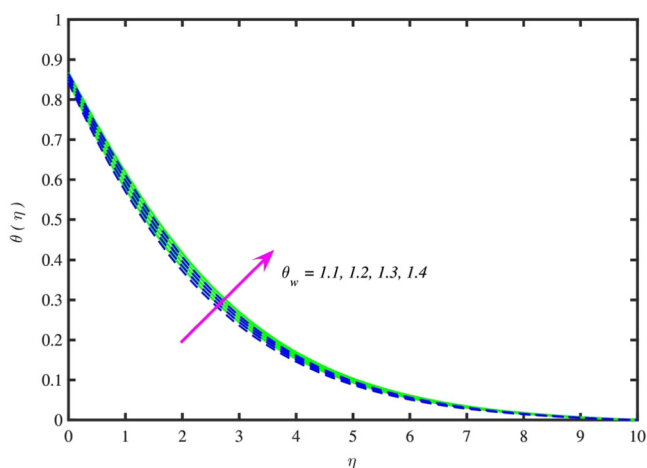


FIGURE 5 Variations of θ_w on temperature [Color figure can be viewed at wileyonlinelibrary.com]

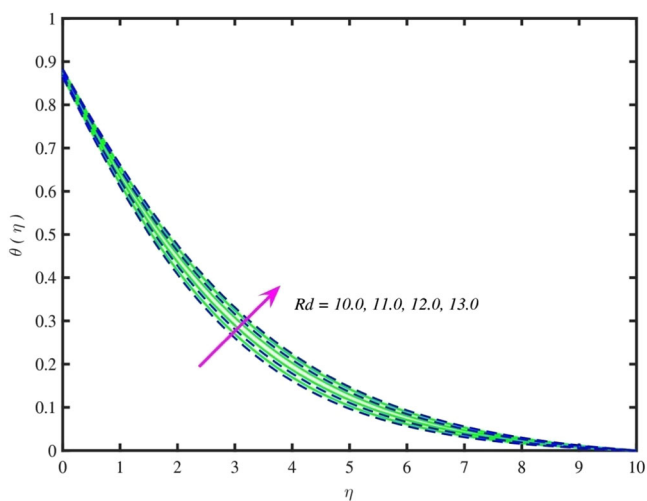


FIGURE 6 Variations of Rd on temperature [Color figure can be viewed at wileyonlinelibrary.com]

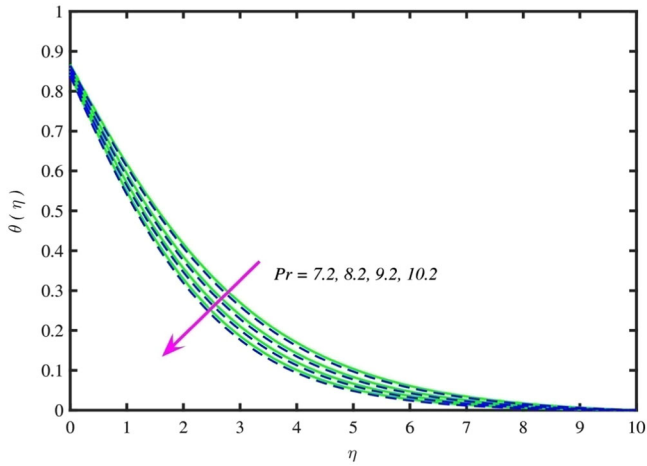


FIGURE 7 Variations of Pr on temperature [Color figure can be viewed at wileyonlinelibrary.com]

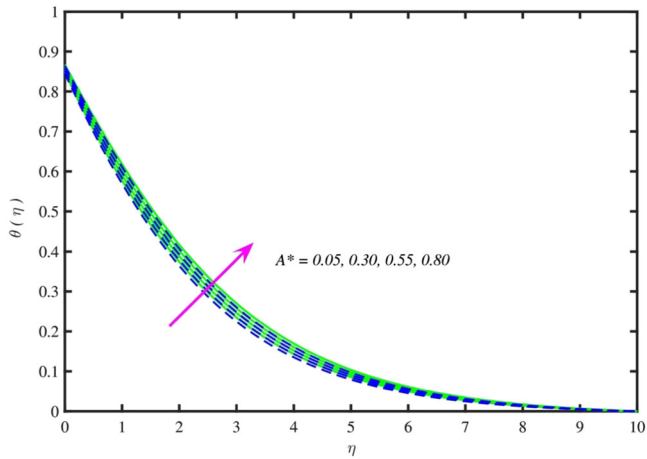


FIGURE 8 Variations of A^* on temperature [Color figure can be viewed at wileyonlinelibrary.com]

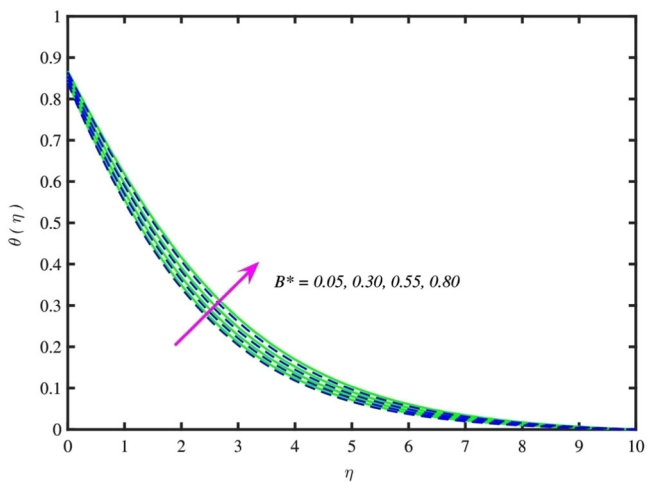


FIGURE 9 Variations of B^* on temperature [Color figure can be viewed at wileyonlinelibrary.com]

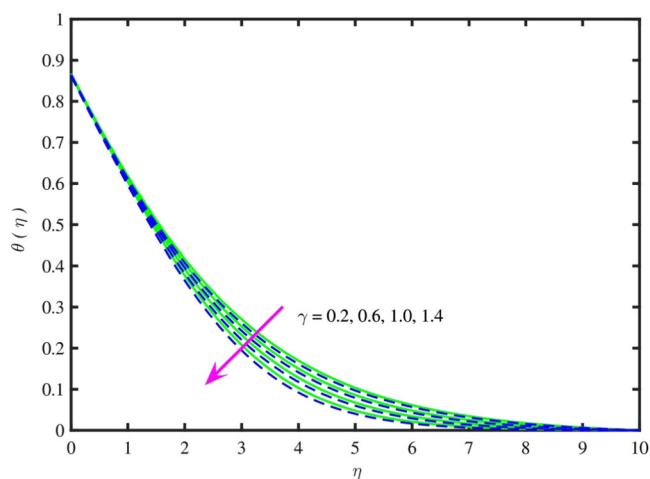


FIGURE 10 Variations of γ on temperature [Color figure can be viewed at wileyonlinelibrary.com]

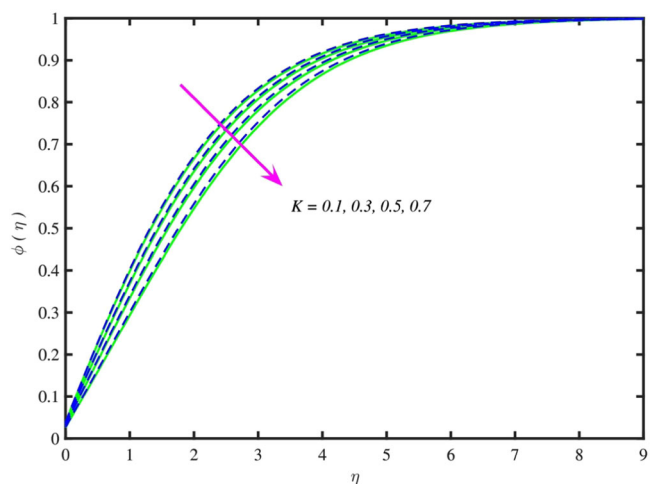


FIGURE 11 Variations of K on concentration [Color figure can be viewed at wileyonlinelibrary.com]

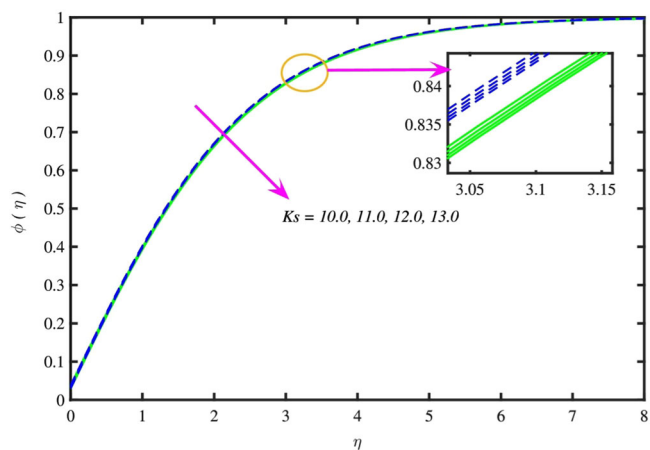


FIGURE 12 Variations of K_s on concentration [Color figure can be viewed at wileyonlinelibrary.com]

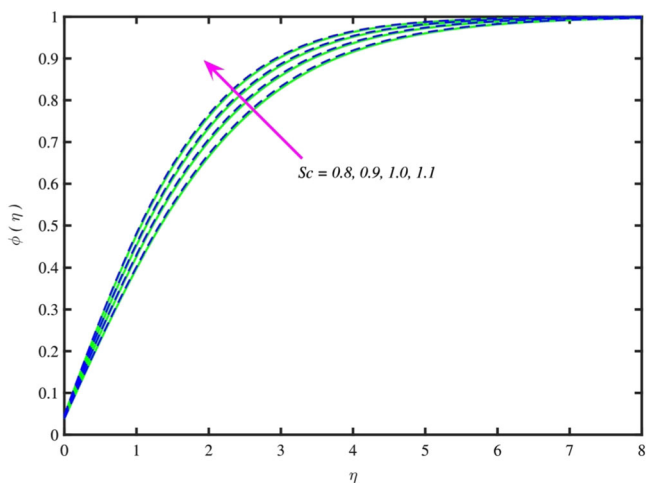


FIGURE 13 Variations of Sc on concentration [Color figure can be viewed at [wileyonlinelibrary.com](#)]

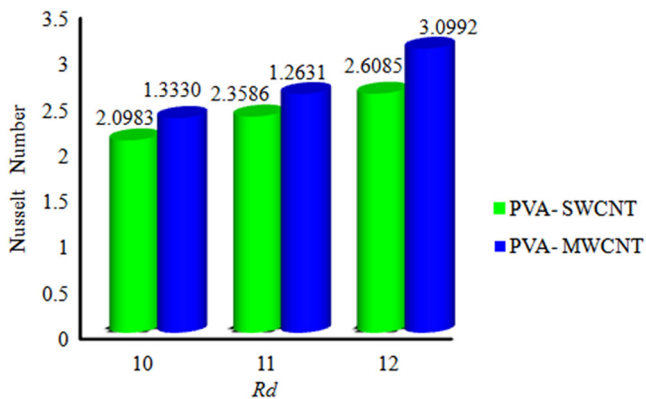


FIGURE 14 Variations of Rd on Nusselt number [Color figure can be viewed at [wileyonlinelibrary.com](#)]

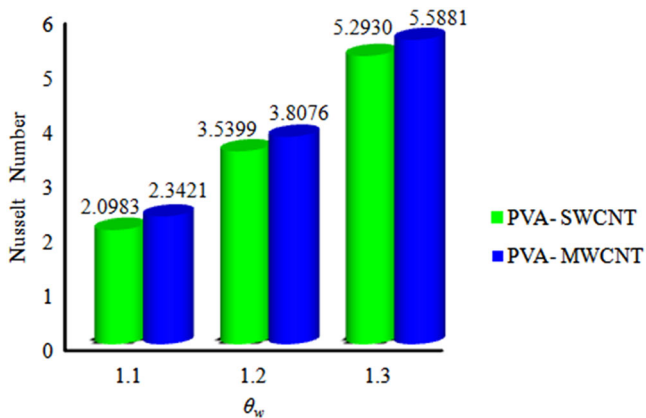


FIGURE 15 Variations of θ_w on Nusselt number [Color figure can be viewed at [wileyonlinelibrary.com](#)]

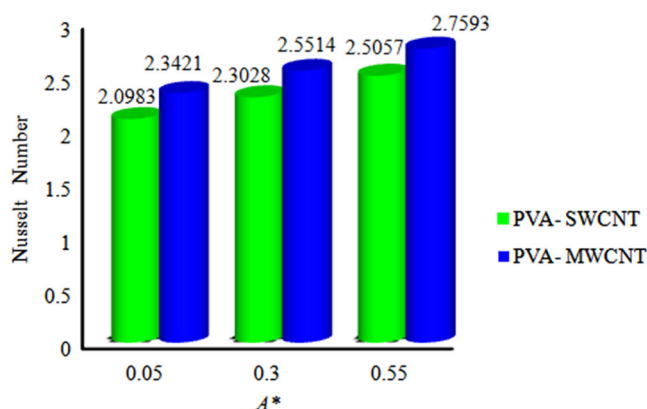


FIGURE 16 Variations of A^* on Nusselt number [Color figure can be viewed at wileyonlinelibrary.com]

4.3 | Effect of thermal slip parameter (ζ)

Figure 4 presents the variation of fluid temperature with thermal slip parameter (ζ). It is obvious that the fluid temperature falls for growing values of ζ . As ζ enhances, less heat is transferred from the stretching plate to the fluid and hence the temperature reduces. We also observe that the thermal boundary layer thickness becomes thicker for mounting values of ζ . A decrement in heat transfer is noticed from sheet to the fluid, as a consequence temperature is reduced. The temperature of SWCNT is slightly more.

4.4 | Effect of temperature ratio parameter (θ_w)

Figure 5 illustrates the variations of θ_w on temperature. From figure it is conveyed that higher estimation of θ_w rises the temperature and also the thickness of interrelated layers. A rise in θ_w yield additional heat to the functioning fluid where a hike in $\theta(\eta)$ is noticed. It is acclaimed that the MWCNTs have dominant temperature profile than SWCNTs.

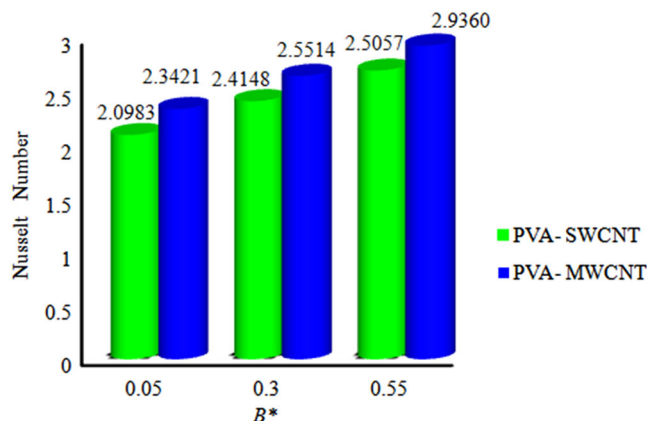


FIGURE 17 Variations of B^* on Nusselt number [Color figure can be viewed at wileyonlinelibrary.com]

4.5 | Effect of nonlinear radiation parameter (rd)

The consequence of Rd on temperature field is conferred in Figure 6. This figure shows that higher assumption of Rd accelerates $\theta(\eta)$. In fact, higher Rd emits more kinetic energy to the process. That is why, temperature field and thermal boundary layer thickness booms for increment in Rd . In the process of radiation additional heat is observed by the functioning fluid which ends in temperature augmentation.

4.6 | Effect of Prandtl number (Pr)

Fluctuation in temperature via Pr is conferred in Figure 7. The temperature field fall for both cases by uplifting the values of Pr . In problems where heat transfer come about, a reduction in the comparative coagulation of the velocity and the thermal boundary layer is noticed by employing Pr . Pr is a sign of the fraction of momentum diffusivity to thermal diffusivity. Fluids with low Pr will acquire high conductivities, so that heat can distribute faster from the sheet than for higher Pr fluids.

4.7 | Effect of space dependent heat source/sink parameter (a^*)

Figure 8 depicts the impression of A^* on temperature. It is witnessed that the boundary layer produces energy, which results in raising the temperature for $A^* > 0$ (heat source) while for $A^* < 0$ (absorption) boundary layer take in energy which results a fall in the temperature. It is noted that the MWCNTs have improved the thermal profile somewhat more than SWCNTs.

4.8 | Effect of temperature-dependent heat source/sink parameter (B^*)

The effect of B^* on temperature is elaborated in Figure 9. This profile demonstrates that temperature raises $B^* > 0$ due to ejection of energy. For $B^* < 0$, the temperature fall nearer to the boundary layer due to the absorption of energy. Also absorbed that the MWCNTs has raised the temperature profile to some extent more than SWCNTs.

4.9 | Effect of thermal relaxation parameter (γ)

To explore the behavior of γ on temperature profile, Figure 10 is plotted. It is given as the impeded reaction of heat flux. With the evidence of the figure, it is examined that the temperature profile inflates by raising the values of γ . The rate of decrement is more for MWCNT, because effective thermal conductivity of SWCNTs is more than the MWCNTs.

4.10 | Effect of homogeneous and heterogeneous reactions (K, K_s)

Profiles of nondimensional concentration profiles are plotted in Figures 11 and 12 for altering magnitude of K and K_s on the concentration. The natural processes of chemical reactions connect mutually with homogeneous and heterogeneous reactions. The relations between homogeneous and heterogeneous reactions are very composite. A number of reactions progress gradually or instantly in absence of a catalyst. Physically homogeneous

variable and chemical reaction are directly related so solutal boundary layer thickness enhances. As K and K_s increases, concentration is gradually decreasing. This is due to more consumption of chemical reactants for higher homogeneous and heterogeneous reactions, which leads a decline in concentration. The decrement of concentration for MWCNTs is more than SWCNTs.

4.11 | Effect of Schmidt number (Sc)

Figure 13 is sketched to interrogate the nanoparticles concentration for different values of Sc . An enhancement of nanoparticles concentration and its related boundary layer thickness accelerates for higher Sc . For a rigid molecular diffusion rate, the viscous diffusion rate increases by increasing Sc , which is proportionate to concentration and inversely to the velocity of the nanofluid.

4.12 | Effect of Nusselt number on nonlinear radiation (rd) and thermal ratio parameter (θ_w)

Figure 14 epitomizes the effect of Nusselt number on Rd for both SWCNTs and MWCNTs. The outcome shows a rise in the Nusselt number for both cases. The effect of temperature ratio parameter θ_w on Nusselt number for both SWCNTs and MWCNTs are demonstrated in Figure 15. It is detected that rising values of temperature ratio parameter raises the local Nusselt number for both CNTs.

4.13 | Effect of Nusselt number on nonuniform heat source (a^*) and heat sink (b^*)

Figures 16 and 17 epitomize the effect of Nusselt number on A^* and B^* for both SWCNTs and MWCNTs. The outcomes show a rise in the Nusselt number for both CNTs.

5 | FINAL CONCLUSIONS

The current communication scrutinizes a theoretical model of PVA with nanoparticles through homogeneous and heterogenous reactions lying on a stretching sheet by means of nonuniform heat source/sink. The numerical elucidations are established by RKF method- bvp4c codes in MATLAB. The strategic upshots of this framework are outlined below:

1. Velocity and magnetic parameter are inverse to each other.
2. Higher estimation of Rd , θ_w , and nonuniform heat source/sink concentrate the temperature distribution.
3. The thickness of thermal boundary layer decreases by mounting Pr and thermal relaxation parameter.
4. The fluid concentration decreases with the increasing values of homogeneous and heterogeneous reactions.
5. Heat transfer rate is high for both temperature ratio parameter and nonlinear thermal radiation.

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NOMENCLATURE

g	acceleration due to gravity
T_{∞}	ambient fluid temperature
C-CHFM	Cattaneo–Christov heat flux model
K	homogeneous reaction strength
K_s	heterogeneous reaction strength
M	magnetic parameter
Pr	Prandtl number
q_r	radiative heat flux
Sc	Schmidt number
T	temperature
k_{CNT}	thermal conductivities of CNTs
k_f	thermal conductivities of the host fluid
k_{nf}	thermal conductivities of the nanofluid
Rd	thermal radiation
B	uniform magnetic field strength,
U_w	velocity at wall
u	velocity components along the x -axis
v	velocity components along y -axis
v_w	wall mass flux
T_w	wall temperature
σ	nondimensional chemical reaction rate constant
θ_w	temperature ratio parameter
μ_f	viscosity of base fluid
μ_{nf}	viscosity of nanofluids
χ	nanoparticles fraction
ρ_f	density of the base fluid
ρ_{CNT}	thermal conductivities of CNTs
$(\rho C_p)_f$	heat capacity of a fluid
$(\rho C_p)_{nf}$	heat capacitance of the nanofluid
$(\rho C_p)_{CNT}$	heat capacity of CNTs
β_f	thermal expansion of the primary fluid
β_{CNT}	thermal expansion of CNTs
ρ_{nf}	density of the nanofluid
σ^{**}	electric conductivity
α_{nf}	thermal diffusivity of nanofluids

ξ	momentum slip
ζ	thermal slip
λ	thermal buoyancy parameter

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